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MIL-M-28787/421
17 May 1991

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

NATO-STD-4156(MODIFIED) CHANNEL CONTROL MODULE

KEY CODE ATK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic NATO-STD-4156(modified) channel control module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 4.863 failures per million operating hours for class I or 17.631 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be ATK.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
A2-A18	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
B1-B22	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc stress	IV	S	----------Module shall pass all remaining tests----------						
D1	Ibd	V	S	-0.024	1.02	-0.024	1.02	-0.024	1.02	mA
D2	Ibd	V	S	-0.024	0.12	-0.024	0.12	-0.024	0.12	mA
D3	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D4	Ibd	V	S	-0.096	0.20	-0.096	0.20	-0.096	0.20	mA
D5-D9	Ibd	V	S	-0.024	0.12	-0.024	0.12	-0.024	0.12	mA
D10-D15	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D16	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D17	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D18	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	mA
D19	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D20-D22	Ibd	V	S	-0.024	1.02	-0.024	1.02	-0.024	1.02	mA
E1-E5	Ios	VI	S	-225.0	-50.0	-225.0	-50.0	-225.0	-50.0	mA
E6	Ios	VI	S	-3.24	-1.1	-3.24	-1.1	-3.24	-1.1	mA
E7	Ios	VI	S	-130.0	-15.0	-130.0	-15.0	-130.0	-15.0	mA
E8,E9	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E10	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E11	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E12	Ios	VI	S	-225.0	-50.0	-225.0	-50.0	-225.0	-50.0	mA
E13-E16	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E17-E19	Ios	VI	S	-225.0	-50.0	-225.0	-50.0	-225.0	-50.0	mA
E20	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E21-E36	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
F1	Iih	VII	X	-1.55	-0.56	-1.55	-0.56	-1.55	-0.56	mA
F2	Iih	VII	X	-1.75	-0.59	-1.75	-0.59	-1.75	-0.59	mA
F3	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA

- See footnotes at end of table.

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TABLE 1. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F4	Iih	VII	X	-4.92	-2.37	-4.92	-2.37	-4.92	-2.37	mA
F5-F9	Iih	VII	X	-1.55	-0.59	-1.55	-0.59	-1.55	-0.59	mA
F10-F15	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F16	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F17	Iih	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F18	Iih	VII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	mA
F19	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F20	Iih	VII	X	-3.15	-0.56	-3.15	-0.56	-3.15	-0.56	mA
F21	Iih	VII	X	-1.55	-0.56	-1.55	-0.56	-1.55	-0.56	mA
F22	Iih	VII	X	-3.15	-0.56	-3.15	-0.56	-3.15	-0.56	mA
F23-F38	Iih	VII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
G1	Iil	VIII	X	-1.55	-0.56	-1.55	-0.56	-1.55	-0.56	mA
G2	Iil	VIII	X	-1.75	-0.59	-1.75	-0.59	-1.75	-0.59	mA
G3	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G4	Iil	VIII	X	-4.92	-2.37	-4.92	-2.37	-4.92	-2.37	mA
G5-G9	Iil	VIII	X	-1.55	-0.59	-1.55	-0.59	-1.55	-0.59	mA
G10-G15	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G16	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G17	Iil	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G18	Iil	VIII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	mA
G19	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G20	Iil	VIII	X	-3.15	-0.56	-3.15	-0.56	-3.15	-0.56	mA
G21	Iil	VIII	X	-1.55	-0.56	-1.55	-0.56	-1.55	-0.56	mA
G22	Iil	VIII	X	-3.15	-0.56	-3.15	-0.56	-3.15	-0.56	mA
G23-G38	Iil	VIII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
H1-H35	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J37	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K1	Iozh	XI	X	0.0	0.25	0.0	0.25	0.0	0.25	mA
K2-K4	Iozh	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
K5-K7	Iozl	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA

- See footnotes at end of table

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
L1-L8	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L9	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L10,L11	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
M	PSFC	XIII	S	8.0	15.0	8.0	15.0	8.0	15.0	uF
N	Idd	XIV	X	582	1398	582	1398	582	1398	mA
P1,P2	Functional	XV	X	----------		----------		See table XV		
Q1,Q2	Dynamic	XVI	S	----------		----------		See table XVI		
Q3,Q4	Dynamic	XVI	Q	----------		----------		See table XVI		
R1	C1/D	XVIII	Q	2	100	2	100	2	100	pF
R2	C1/D	XVIII	Q	4	100	4	100	4	100	pF
R3	C1/D	XVIII	Q	2	200	2	200	2	200	pF
R4	C1/D	XVIII	Q	4	100	4	100	4	100	pF
R5	C1/D	XVIII	Q	14	100	14	100	14	100	pF
R6	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R7	C1/D	XVIII	Q	4	50	4	50	4	50	pF
R8-R10	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R11-R17	C1/D	XVIII	Q	2	100	2	100	2	100	pF
R18	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R19	C1/D	XVIII	Q	4	100	4	100	4	100	pF
R20	C1/D	XVIII	Q	2	125	2	125	2	125	pF
R21	C1/D	XVIII	Q	2	100	2	100	2	100	pF
R22	C1/D	XVIII	Q	4	125	4	125	4	125	pF
R23-R26	C1/D	XVIII	Q	4	100	4	100	4	100	pF
R27	C1/D	XVIII	Q	2	125	2	125	2	125	pF
R28	C1/D	XVIII	Q	4	125	4	125	4	125	pF
R29-R44	C1/D	XVIII	Q	4	75	4	75	4	75	pF
T	PSTA	XIX	Q	0	400	0	400	0	400	mV

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/421-1 for class I and M28787/421-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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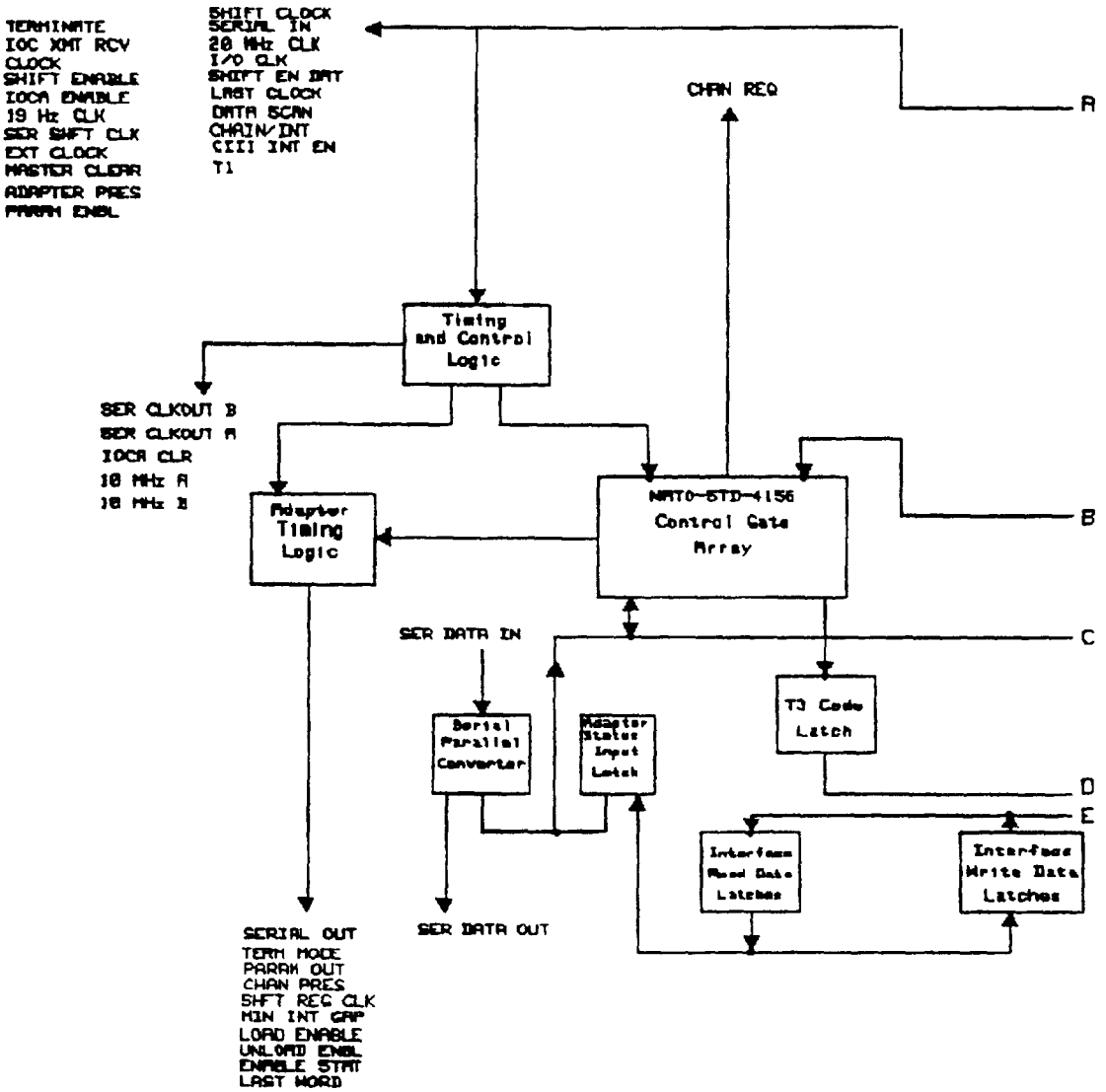


FIGURE 1a. ATK functional block diagram.

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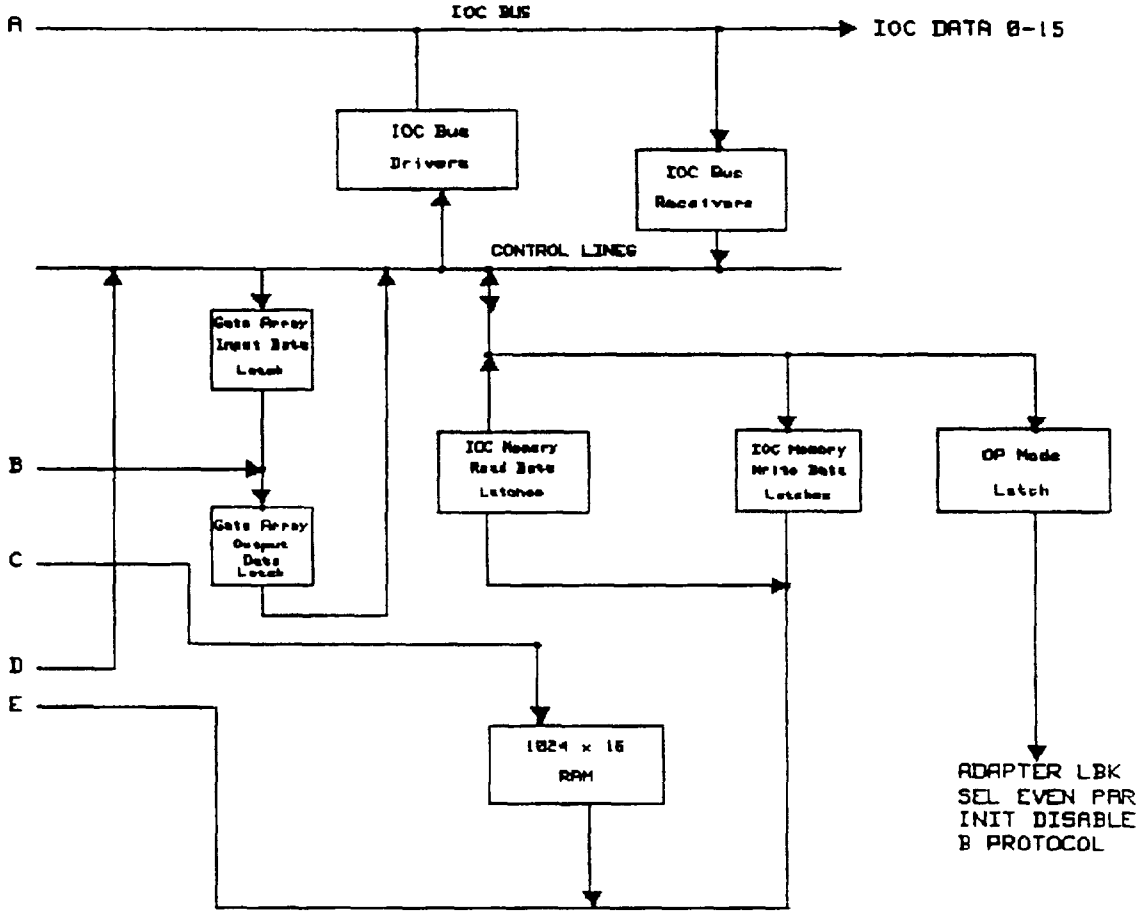


FIGURE 1b. ATK functional block diagram.

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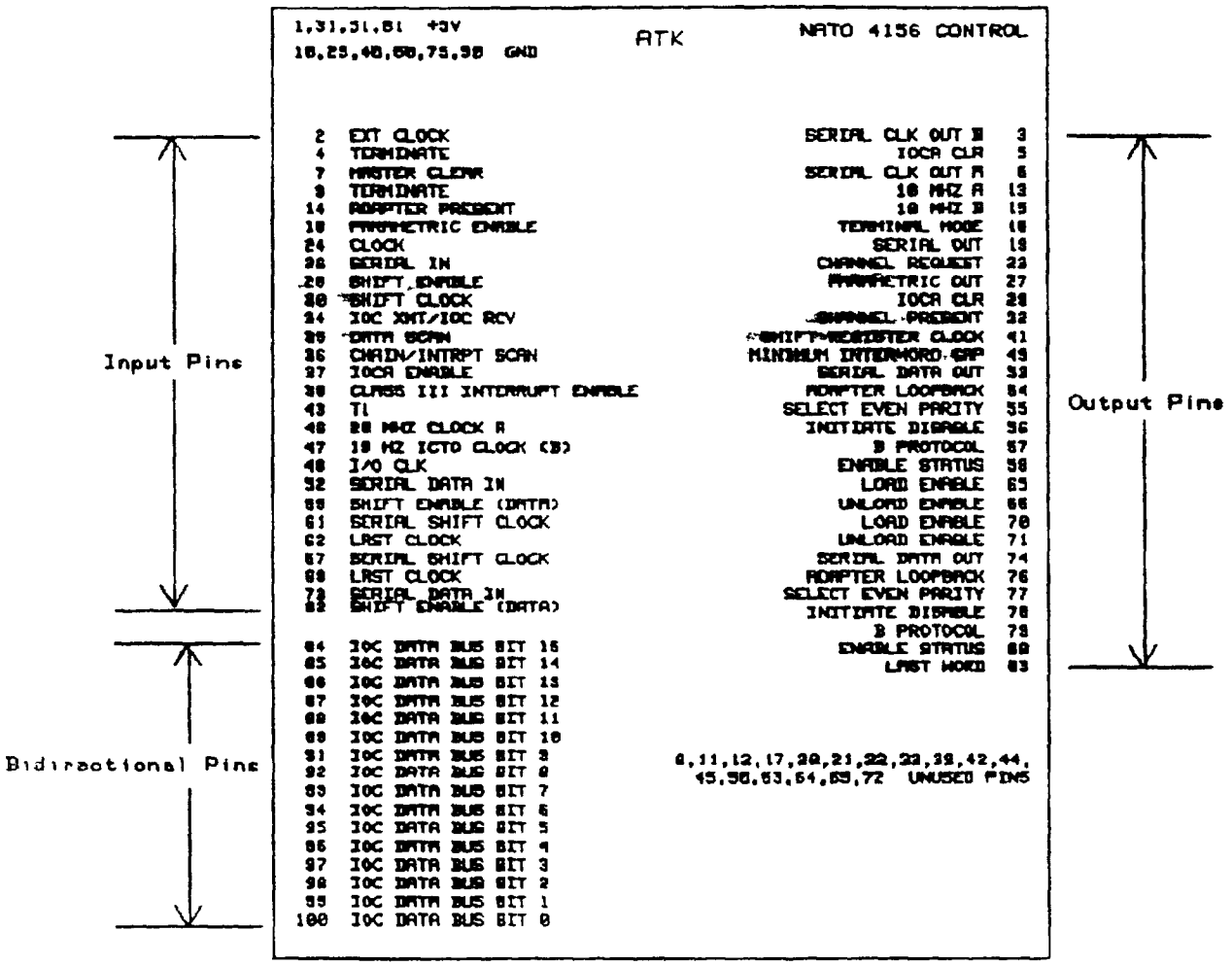


FIGURE 2. ATK pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

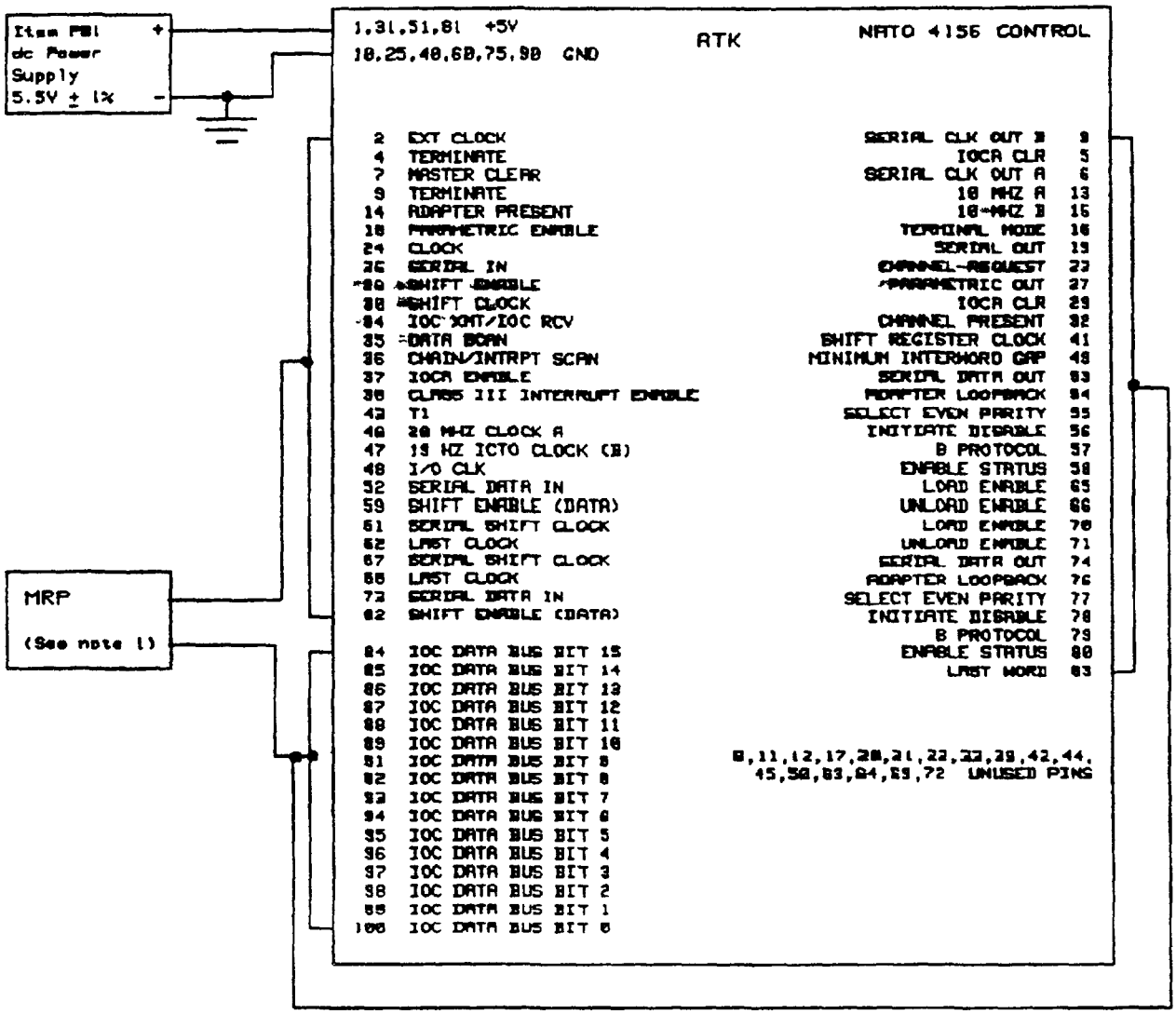
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. ATK operating temperature and component temperature test circuit.

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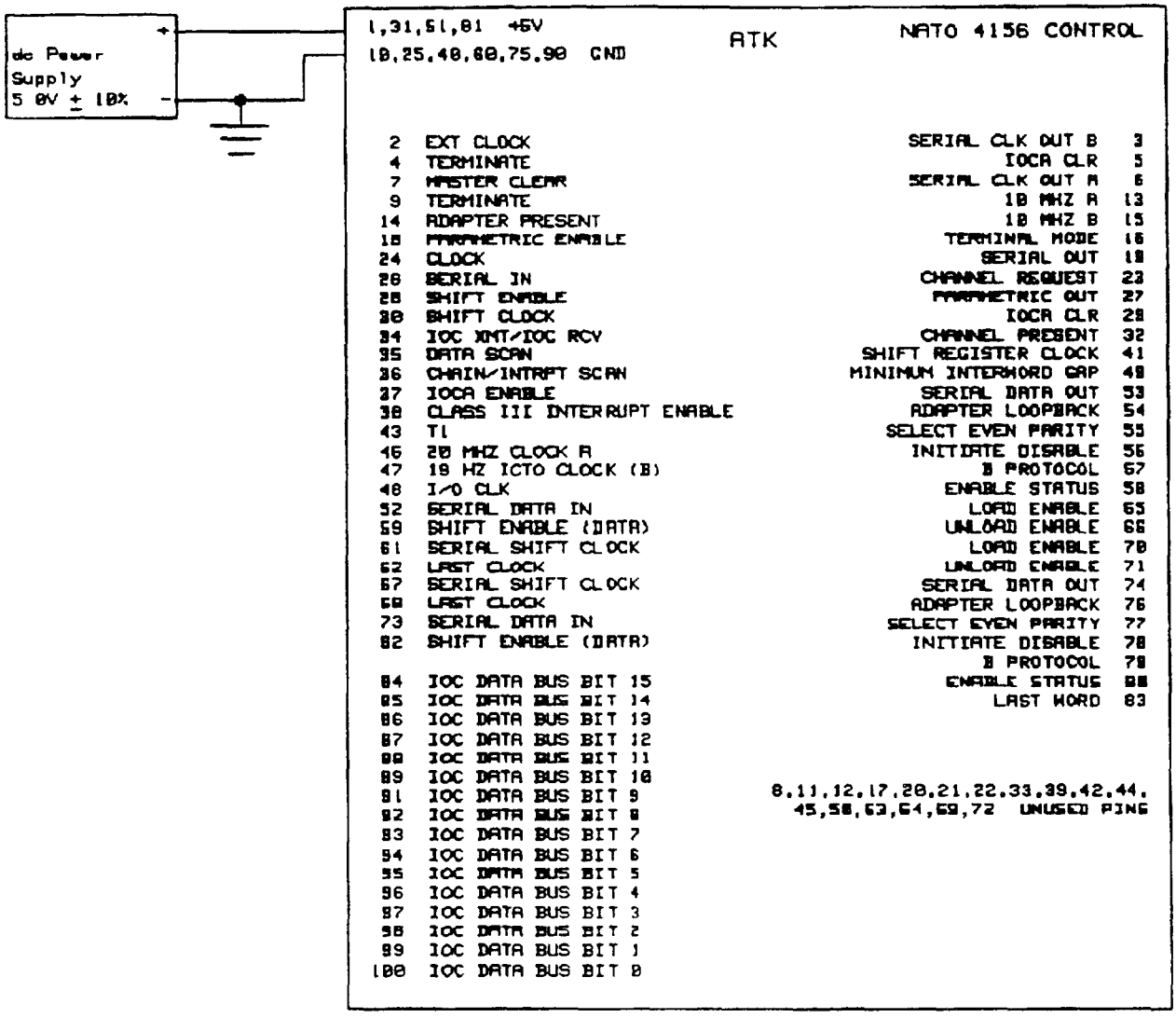


FIGURE 4. ATK vibration test circuit.

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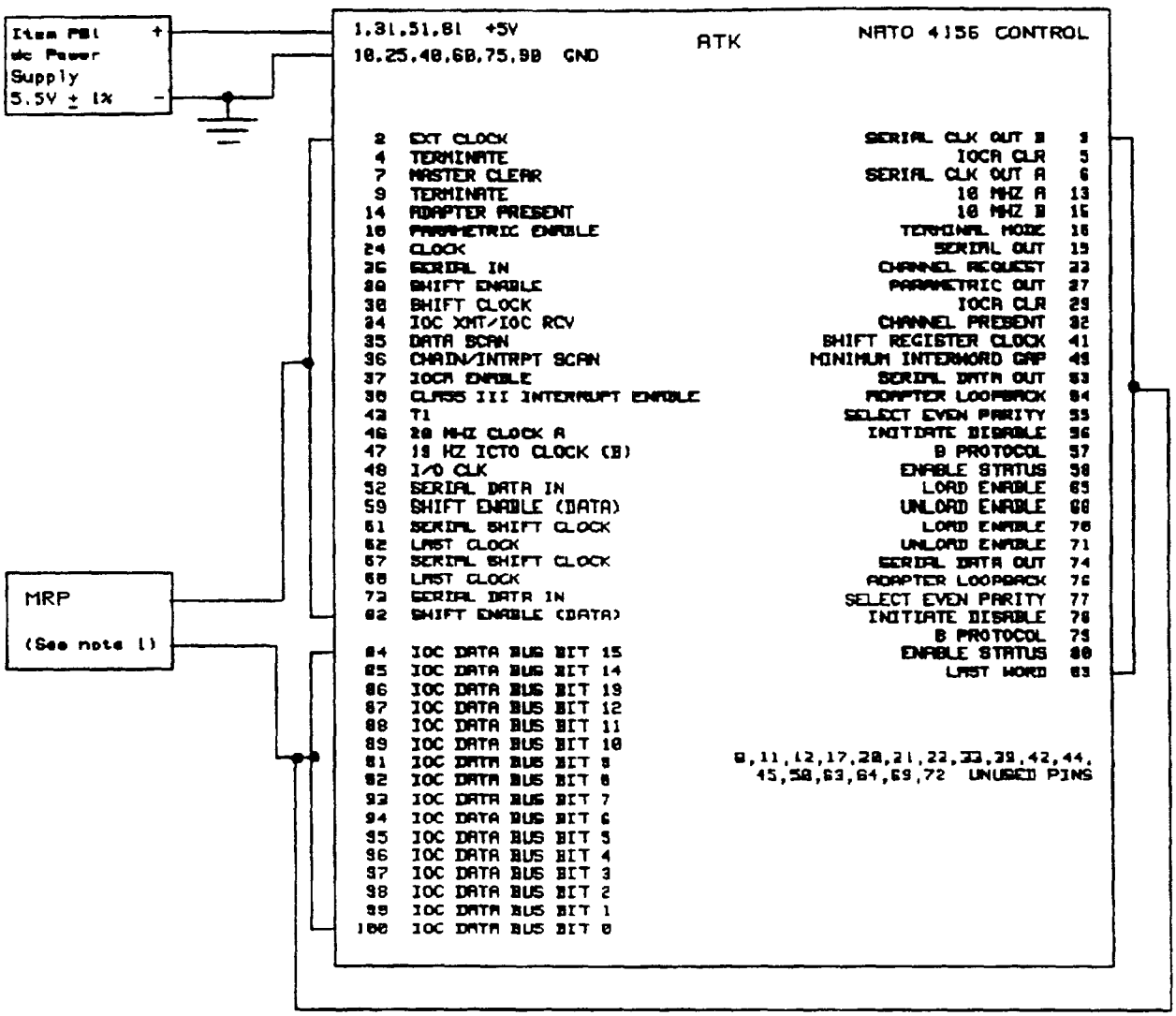
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. ATK life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 7.689 W (max); 5.154 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
EXT CLOCK	NA	0.4	2.4	-1.55	-0.56	1.02	100
TERMINATE	NA	0.4	2.4	-1.75	-0.59	0.12	100
MASTER CLEAR	-1.2	0.4	2.4	-0.6	0.02	0.1	200
ADAPTER PRES	NA	0.4	2.4	-4.92	-2.37	0.20	100
PARAM ENBL	NA	0.4	2.4	-1.55	-0.59	0.12	75
CLOCK	NA	0.4	2.4	-1.55	-0.59	0.12	50
SERIAL IN	NA	0.4	2.4	-1.55	-0.59	0.12	75
SHIFT ENABLE	NA	0.4	2.4	-1.55	-0.59	0.12	75
SHIFT CLOCK	NA	0.4	2.4	-1.55	-0.59	0.12	75
IOC XMT RCV	-1.2	0.4	2.4	-0.6	0.02	0.1	100
DATA SCAN	-1.2	0.4	2.4	-0.6	0.02	0.1	100
CHAIN/INT	-1.2	0.4	2.4	-0.6	0.02	0.1	100
IOCA ENABLE	-1.2	0.4	2.4	-0.6	0.02	0.1	100
CIII INT EN	-1.2	0.4	2.4	-0.6	0.02	0.1	100
T1	-1.2	0.4	2.4	-0.6	0.02	0.1	100

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
20 MHZ CLK	-1.2	0.4	2.4	-2.0	0.05	1.0	100
19 HZ CLK	-1.5	0.4	2.4	-0.4	0.02	0.1	75
I/O CLK	-1.2	0.4	2.4	-4.0	0.1	2.0	100
SER DATA IN	-1.2	0.4	2.4	-2.0	0.05	1.0	125
SHIFT EN DAT	NA	0.4	2.4	-3.15	-0.56	1.02	125
SER SHFT CLK	NA	0.4	2.4	-1.55	-0.56	1.02	100
LAST CLOCK	NA	0.4	2.4	-3.15	-0.56	1.02	100
IOC DATA 15	NA	0.4	2.4	-0.25	0.07	NA	75
IOC DATA 14	NA	0.4	2.4	-0.25	0.07	NA	75
IOC DATA 13	NA	0.4	2.4	-0.25	0.07	NA	75
IOC DATA 12	NA	0.4	2.4	-0.25	0.07	NA	75
IOC DATA 11	NA	0.4	2.4	-0.25	0.07	NA	75
IOC DATA 10	NA	0.4	2.4	-0.25	0.07	NA	75
IOC DATA 9	NA	0.4	2.4	-0.25	0.07	NA	75
IOC DATA 8	NA	0.4	2.4	-0.25	0.07	NA	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
IOC DATA 7	NA	0.4	2.4	-0.25	0.07	NA	75
IOC DATA 6	NA	0.4	2.4	-0.25	0.07	NA	75
IOC DATA 5	NA	0.4	2.4	-0.25	0.07	NA	75
IOC DATA 4	NA	0.4	2.4	-0.25	0.07	NA	75
IOC DATA 3	NA	0.4	2.4	-0.25	0.07	NA	75
IOC DATA 2	NA	0.4	2.4	-0.25	0.07	NA	75
IOC DATA 1	NA	0.4	2.4	-0.25	0.07	NA	75
IOC DATA 0	NA	0.4	2.4	-0.25	0.07	NA	75

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh min (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
SER CLKOUT B	2.4	0.5	-0.4	6.0	NA	NA	-50.0	NA
IOCA CLR	2.4	0.5	-0.4	6.0	NA	NA	-50.0	NA
SER CLKOUT A	2.4	0.5	-0.4	6.0	NA	NA	-50.0	NA
10 MHZ A	2.4	0.5	-0.4	6.0	NA	NA	-50.0	NA
10 MHZ B	2.4	0.5	-0.4	6.0	NA	NA	-50.0	NA
TERM MODE	2.4	0.5	-0.16	1.6	NA	NA	-1.1	NA
SERIAL OUT	2.4	0.5	-0.16	1.6	NA	NA	-15.0	NA
CHAN REQ	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
PARAM OUT	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
CHAN PRES	2.4	0.5	-0.16	1.6	0.25	NA	NA	NA
SHFT REG CLK	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
MIN INT GAP	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
SER DATA OUT	2.4	0.5	-0.4	6.0	NA	NA	-50.0	100
ADAPTER LBK	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
SEL EVEN PAR	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh min (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
INIT DISABLE	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
B PROTOCOL	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
ENABLE STAT	2.4	0.5	-0.4	6.0	0.05	0.05	-50.0	NA
UNLOAD ENBL	2.4	0.5	-0.4	6.0	0.05	0.05	-50.0	NA
LOAD ENABLE	2.4	0.5	-0.4	6.0	0.05	0.05	-50.0	NA
LAST WORD	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
IOC DATA 15	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC DATA 14	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC DATA 13	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC DATA 12	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC DATA 11	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC DATA 10	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC DATA 9	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC DATA 8	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC DATA 7	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh min (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
IOC DATA 6	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC DATA 5	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC DATA 4	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC DATA 3	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC DATA 2	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC DATA 1	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC DATA 0	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	MASTER CLEAR to IOCA CLR	-	-	-	55	55	55	nS	11
2	tpd	I/O CLK to TERM MODE	-	-	-	109	109	109	nS	11
3	tpd	DATA SCAN to CHAN REQ	-	-	-	48	48	48	nS	11
4	tpd	20 MHZ CLK to LOAD ENABLE	-	-	-	104	104	104	nS	11
5	tpd	I/O CLK to INIT DISABLE	-	-	-	57	57	57	nS	11
6	tpd	I/O CLK to IOC DATA 0	-	-	-	47	47	47	nS	11
7	tpd	SER SHFT CLK to SER DATA OUT	-	-	-	58	58	58	nS	11

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6.4 Module function. The NATO-STD-4156 (modified) channel control module contains the circuits necessary to direct the operation of two NATO-STD-4156 Serial Input/Output (I/O) channels. The module contains timing and control logic, a NATO-STD-4156 serial channel control gate array, adapter timing logic, T3 code latch, serial-parallel converter, interface read/write data latches, Input/Output Controller (IOC) bus drivers and receivers, gate array input/output data latches, 1024 x 16-bit Random Access Memory (RAM), IOC memory read/write data latches and an Op mode latch.

6.4.1 Control gate array. The control gate array performs I/O channel-related functions and AN/UYK-44 IOC bus interface and data transfer control operations.

6.4.2 IOC bus drivers and receivers. IOC bus receivers direct the flow of data from the AN/UYK-44 IOC bus to the Input/ Output Channel Adapter (IOCA) control lines. The receivers are controlled by the IOC transmit/receive signal. The IOC bus drivers direct the flow of data from the IOCA control lines to the IOC bus. The drivers are controlled by the IOC transmit/receive signal and the IOCA enable.

6.4.3 Op mode latch. The op mode latch holds bits 14, 12, 10 and 8 of the serial mode word which are loaded from the control lines by the load op mode signal from the NATO-STD-4156 control gate array. Outputs are sent to the NATO-STD-4156 serial channel adapter (ATL).

6.4.4 IOCA memory read/write data latches. The read/write data latches hold data read from and data to be written into the 1024 x 16-bit RAM (buffer memory). Data is clocked into and out of these latches by the control gate array.

6.4.5 1024 x 16 RAM. The 1024-word by 16-bit RAM is the IOCA buffer memory which can hold data to or from the IOC, and control words from the gate array.

6.4.6 Gate array input/output data latches. The gate array input/output data latches hold commands and data to and from the IOC and are clocked by the timing and control logic.

6.4.7 T3 code latch. The T3 code latch holds the selected control data which is output to the IOC bus at T3 time. The latch is enabled by the T3 timing signal.

6.4.8 Interface read/write data latches. The interface data read/write latches are 16-bit latches that hold data sent from or data destined for the interface modules.

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6.4.9 Adapter status input latch. The adapter status input latch holds the 16-bit I/O status word. The latch is loaded and enabled by the enable status signal.

6.4.10 Serial-parallel converter. The serial to parallel converter is composed of two 8-bit shift registers, and is used to convert the parallel data from the control module to serial output data, and convert the serial input data to parallel data on the control module.

6.4.11 Timing and control logic. The timing and control logic generates the timing required by the control gate array from the IOC bus control signals.

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6.4.12 Description of I/O pins.

Signal name	Pin number	I/O	Description
EXT CLOCK	2	I	EXT CLOCK
TERMINATE	4	I	TERMINATE
MASTER CLEAR	7	I	MASTER CLEAR
TERMINATE	9	I	TERMINATE
ADAPTER PRES	14	I	ADAPTER PRESENT
PARAM ENBL	18	I	PARAMETRIC ENABLE
CLOCK	24	I	CLOCK
SERIAL IN	26	I	SERIAL IN
SHIFT ENABLE	28	I	SHIFT ENABLE
SHIFT CLOCK	30	I	SHIFT CLOCK
IOC XMT RCV	34	I	IOC XMT/IOC RCV
DATA SCAN	35	I	DATA SCAN
CHAIN/INT	36	I	CHAIN/INTRPT SCAN
IOCA ENABLE	37	I	IOCA ENABLE
CIII INT EN	38	I	CLASS III INTERRUPT ENABLE
T1	43	I	T1
20 MHZ CLK	46	I	20 MHZ CLOCK A
19 HZ CLK	47	I	19 HZ ICTO CLOCK [B]
I/O CLK	48	I	I/O CLOCK
SER DATA IN	52	I	SERIAL DATA IN

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6.4.12 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
SHIFT EN DAT	59	I	SHIFT ENABLE [DATA]
SER SHFT CLK	61	I	SERIAL SHIFT CLOCK
LAST CLOCK	62	I	LAST CLOCK
SER SHFT CLK	67	I	SERIAL SHIFT CLOCK
LAST CLOCK	68	I	LAST CLOCK
SER DATA IN	73	I	SERIAL DATA IN
SHIFT EN DAT	82	I	SHIFT ENABLE [DATA]
SER CLKOUT B	3	O	SERIAL CLK OUT B
IOCA CLR	5	O	IOCA CLR
SER CLKOUT A	6	O	SERIAL CLK OUT A
10 MHZ A	13	O	10 MHZ A
10 MHZ B	15	O	10 MHZ B
TERM MODE	16	O	TERMINAL MODE
SERIAL OUT	19	O	SERIAL OUT
CHAN REQ	23	O	CHANNEL REQUEST
PARAM OUT	27	O	PARAMETRIC OUT
IOCA CLR	29	O	IOCA CLR
CHAN PRES	32	O	CHANNEL PRESENT
SHFT REG CLK	41	O	SHIFT REGISTER CLOCK
MIN INT GAP	49	O	MINIMUM INTERWORD GAP

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6.4.12 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
SER DATA OUT	53	O	SERIAL DATA OUT
ADAPTER LBK	54	O	ADAPTER LOOPBACK
SEL EVEN PAR	55	O	SELECT EVEN PARITY
INIT DISABLE	56	O	INITIATE DISABLE
B PROTOCOL	57	O	B PROTOCOL
ENABLE STAT	58	O	ENABLE STATUS
LOAD ENABLE	65	O	LOAD ENABLE
UNLOAD ENBL	66	O	UNLOAD ENABLE
LOAD ENABLE	70	O	LOAD ENABLE
UNLOAD ENBL	71	O	UNLOAD ENABLE
SER DATA OUT	74	O	SERIAL DATA OUT
ADAPTER LBK	76	O	ADAPTER LOOPBACK
SEL EVEN PAR	77	O	SELECT EVEN PARITY
INIT DISABLE	78	O	INITIATE DISABLE
B PROTOCOL	79	O	B PROTOCOL
ENABLE STAT	80	O	ENABLE STATUS
LAST WORD	83	O	LAST WORD
IOC DATA 15	84	I/O	IOC DATA BUS BIT 15
IOC DATA 14	85	I/O	IOC DATA BUS BIT 14
IOC DATA 13	86	I/O	IOC DATA BUS BIT 13

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6.4.12 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
IOC DATA 12	87	I/O	IOC DATA BUS BIT 12
IOC DATA 11	88	I/O	IOC DATA BUS BIT 11
IOC DATA 10	89	I/O	IOC DATA BUS BIT 10
IOC DATA 9	91	I/O	IOC DATA BUS BIT 9
IOC DATA 8	92	I/O	IOC DATA BUS BIT 8
IOC DATA 7	93	I/O	IOC DATA BUS BIT 7
IOC DATA 6	94	I/O	IOC DATA BUS BIT 6
IOC DATA 5	95	I/O	IOC DATA BUS BIT 5
IOC DATA 4	96	I/O	IOC DATA BUS BIT 4
IOC DATA 3	97	I/O	IOC DATA BUS BIT 3
IOC DATA 2	98	I/O	IOC DATA BUS BIT 2
IOC DATA 1	99	I/O	IOC DATA BUS BIT 1
IOC DATA 0	100	I/O	IOC DATA BUS BIT 0

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0074)

Agent: NW

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	C,8, 11,12, 17,20, 21,22, 33,39, 42,44, 45,50, 63,64, 69,72	Frame	10	INF	MOhm	1-9
A2	Isolation, used to unused pins	C	8,11, 12,17, 20,21, 22,33, 39,42, 44,45, 50,63, 64,69, 72	10	INF	MOhm	1-9
A3-A18	Isolation, unused to unused pins	see note 4		10	INF	MOhm	1-9

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1-A18
Isolation measurement	NA	± 500 KOhm	A1-A18

2. The ohmmeter used shall be item R06.

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NOTES: - Continued.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. The unused pins on this module are: 8, 11, 12, 17, 20, 21, 22, 33, 39, 42, 44, 45, 50, 63, 64, 69, 72.

5. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

6. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

7. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

8. The test circuit shall be as indicated on figure 6.

9. Abbreviations used are defined in 6.1.

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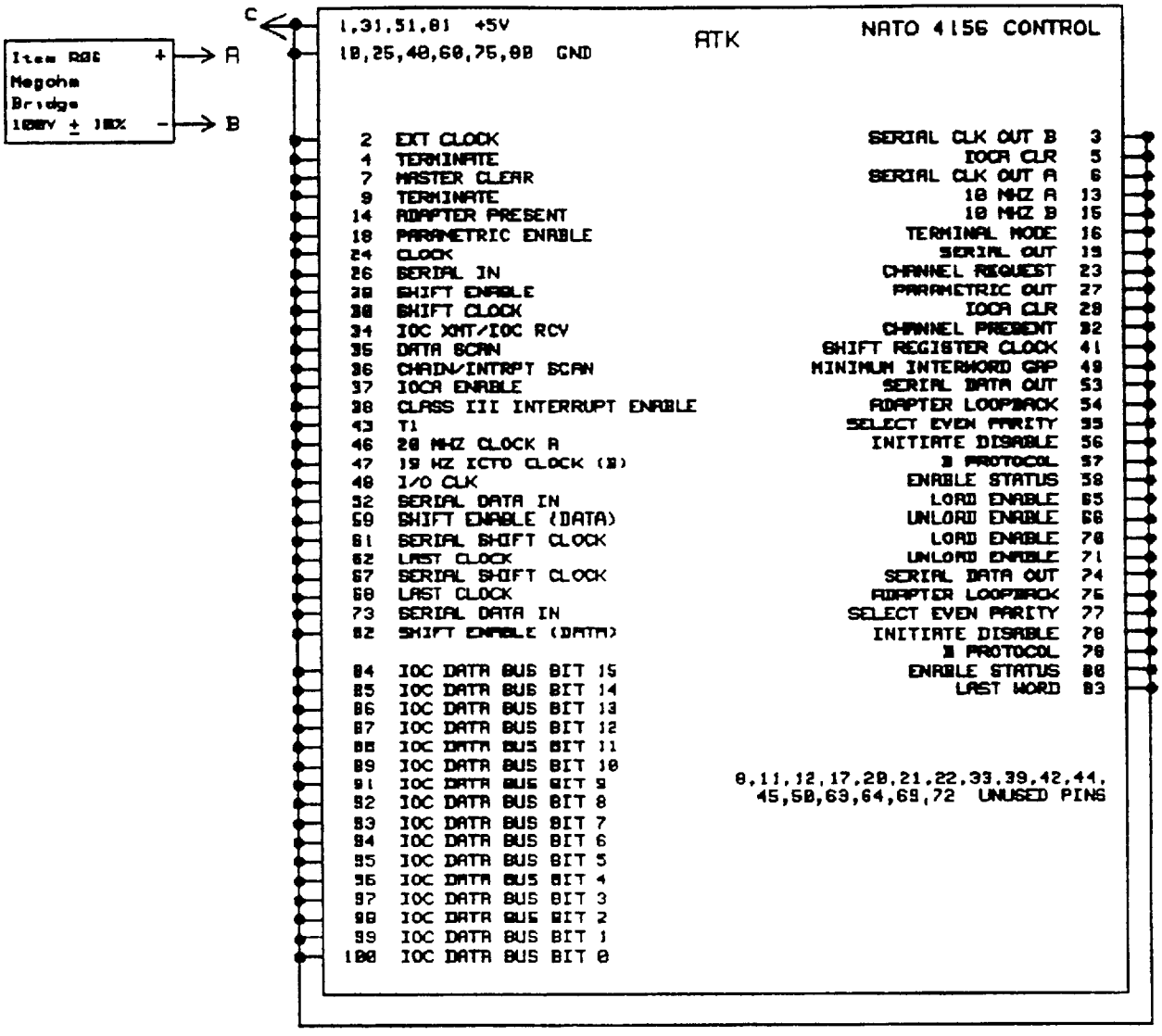


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	GND	10	25	0	1	Ohm	1-5
B2	GND	10	40				
B3	GND	10	60				
B4	GND	60	75				
B5	GND	60	90				
B6	+5V	1	31				
B7	+5V	1	51				
B8	+5V	51	81				
B9	IOCA CLR	5	29				
B10	SER DATA OUT	53	74				
B11	ADAPTER LBK	54	76				
B12	SEL EVEN PAR	55	77				
B13	INIT DISABLE	56	78				
B14	B PROTOCOL	57	79				
B15	ENABLE STAT	58	80				
B16	LOAD ENABLE	65	70				
B17	UNLOAD ENBL	66	71				
B18	TERMINATE	4	9				
B19	SHIFT EN DAT	59	82				
B20	SER SHFT CLK	61	67				
B21	LAST CLOCK	62	68				
B22	SER DATA IN	52	73	0	1	Ohm	1-5

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NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B22

2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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1,31,51,81 +5V
10,25,40,60,75,90 GND

ATK

NATO 4156 CONTROL

2	EXT CLOCK	SERIAL CLK OUT B	3
4	TERMINATE	IOCR CLR	5
7	MASTER CLEAR	SERIAL CLK OUT A	6
8	TERMINATE	10 MHZ A	13
14	ADAPTER PRESENT	10 MHZ B	15
18	PARAMETRIC ENABLE	TERMINAL MODE	16
24	CLOCK	SERIAL OUT	19
26	SERIAL IN	CHANNEL REQUEST	23
28	SHIFT ENABLE	PARAMETRIC OUT	27
30	SHIFT CLOCK	IOCR CLR	29
34	IOC XMT/IOC RCV	CHANNEL PRESENT	32
35	DATA SCAN	SHIFT REGISTER CLOCK	41
36	CHAIN/INTRPT SCAN	MINIMUM INTERWORD GAP	49
37	IOCA ENABLE	SERIAL DATA OUT	53
38	CLASS III INTERRUPT ENABLE	ADAPTER LOOPBACK	54
43	T1	SELECT EVEN PARITY	55
46	20 MHZ CLOCK A	INITIATE DISABLE	56
47	19 HZ ICTO CLOCK (B)	B PROTOCOL	57
48	I/O CLK	ENABLE STATUS	58
52	SERIAL DATA IN	LOAD ENABLE	65
59	SHIFT ENABLE (DATA)	UNLOAD ENABLE	66
61	SERIAL SHIFT CLOCK	LOAD ENABLE	70
62	LAST CLOCK	UNLOAD ENABLE	71
67	SERIAL SHIFT CLOCK	SERIAL DATA OUT	74
68	LAST CLOCK	ADAPTER LOOPBACK	76
73	SERIAL DATA IN	SELECT EVEN PARITY	77
82	SHIFT ENABLE (DATA)	INITIATE DISABLE	78
		B PROTOCOL	79
84	IOC DATA BUS BIT 15	ENABLE STATUS	80
85	IOC DATA BUS BIT 14	LAST WORD	83
86	IOC DATA BUS BIT 13		
87	IOC DATA BUS BIT 12		
88	IOC DATA BUS BIT 11		
89	IOC DATA BUS BIT 10		
91	IOC DATA BUS BIT 9	8,11,12,17,20,21,22,33,39,42,44,	
92	IOC DATA BUS BIT 8	45,50,63,64,69,72 UNUSED PINS	
93	IOC DATA BUS BIT 7		
94	IOC DATA BUS BIT 6		
95	IOC DATA BUS BIT 5		
96	IOC DATA BUS BIT 4		
97	IOC DATA BUS BIT 3		
98	IOC DATA BUS BIT 2		
99	IOC DATA BUS BIT 1		
100	IOC DATA BUS BIT 0		

FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	2.516 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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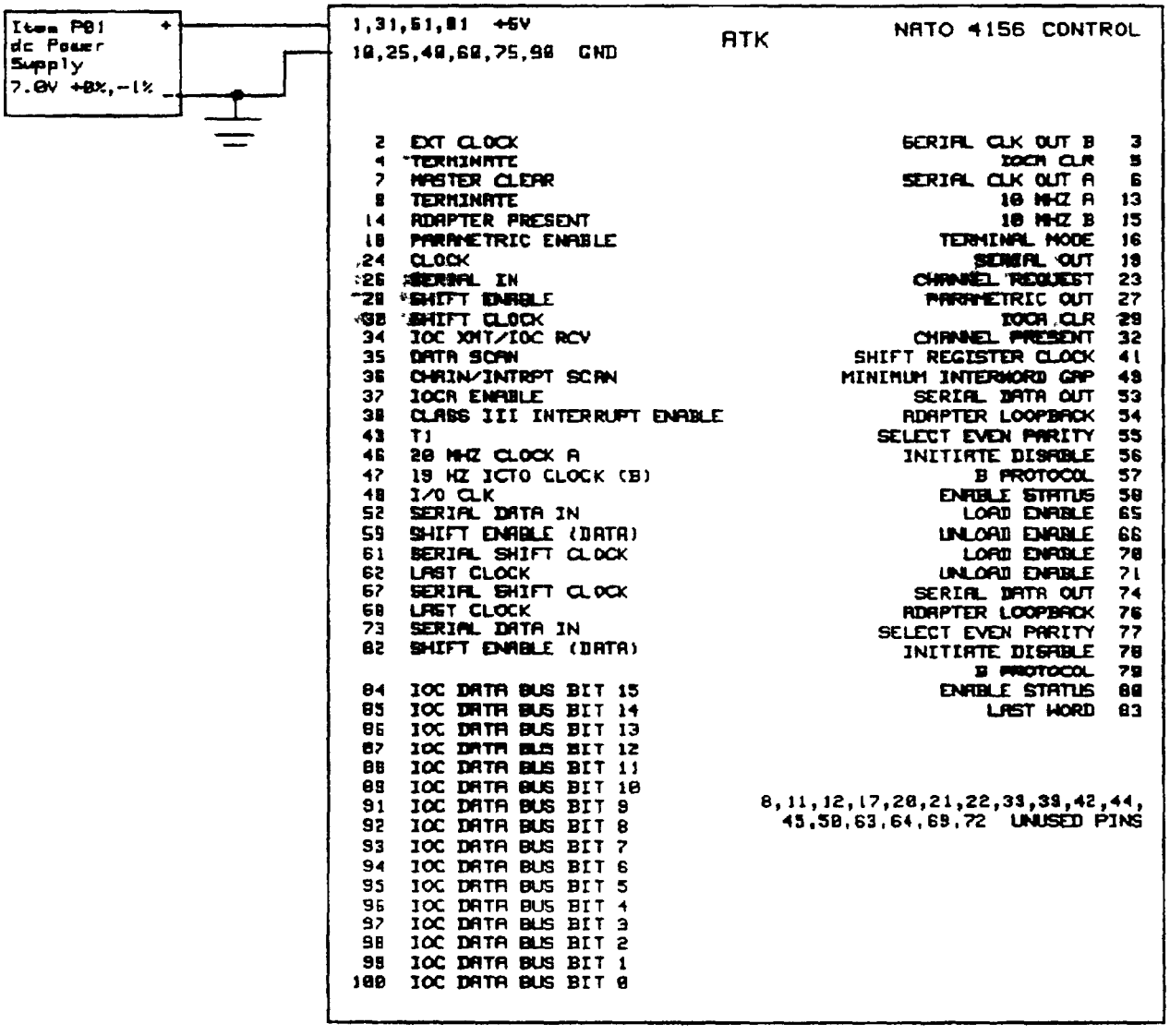


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	EXT CLOCK	2	5.5	-0.024	1.02	mA	1-3
D2	TERMINATE	4	5.5	-0.024	0.12	mA	1-3
D3	MASTER CLEAR	7	5.5	0.0	0.1	mA	1-3
D4	ADAPTER PRES	14	5.5	-0.096	0.20	mA	1-3
D5	PARAM ENBL	18	5.5	-0.024	0.12	mA	1-3
D6	CLOCK	24					
D7	SERIAL IN	26					
D8	SHIFT ENABLE	28					
D9	SHIFT CLOCK	30	5.5	-0.024	0.12	mA	1-3
D10	IOC XMT RCV	34	5.5	0.0	0.1	mA	1-3
D11	DATA SCAN	35					
D12	CHAIN/INT	36					
D13	IOCA ENABLE	37					
D14	CIII INT EN	38					
D15	T1	43	5.5	0.0	0.1	mA	1-3
D16	20 MHZ CLK	46	5.5	0.0	1.0	mA	1-3
D17	19 HZ CLK	47	5.5	0.0	0.1	mA	1-3
D18	I/O CLK	48	5.5	0.0	2.0	mA	1-3
D19	SER DATA IN	52	5.5	0.0	1.0	mA	1-3
D20	SHIFT EN DAT	59	5.5	-0.024	1.02	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	SER SHFT CLK	61	5.5	-0.024	1.02	mA	1-3
D22	LAST CLOCK	62	5.5	-0.024	1.02	mA	1-3

NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D22
Power supply current limit	2.1 A	NA	D1-D22
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D22
Pattern voltage high, (VH)	3.0 V	± 100 mV	D1-D22
low, (VL)	0.0 V	± 100 mV	D1-D22
Input current measurement	NA	± 52.2 uA	D1
	NA	± 7.2 uA	D2
	NA	± 5.0 uA	D3
	NA	± 14.8 uA	D4
	NA	± 7.2 uA	D5-D9
	NA	± 5.0 uA	D10-D15
	NA	± 50.0 uA	D16
	NA	± 5.0 uA	D17
	NA	±100.0 uA	D18
	NA	± 50.0 uA	D19
	NA	± 52.2 uA	D20-D22

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	SER CLKOUT B	3	0.0	-225.0	-50.0	mA	1-4
E2	IOCA CLR	5					
E3	SER CLKOUT A	6					
E4	10 MHZ A	13					
E5	10 MHZ B	15	0.0	-225.0	-50.0	mA	1-4
E6	TERM MODE	16	0.0	-3.24	-1.1	mA	1-4
E7	SERIAL OUT	19	0.0	-130.0	-15.0	mA	1-4
E8	CHAN REQ	23	0.0	-150.0	-60.0	mA	1-4
E9	PARAM OUT	27	0.0	-150.0	-60.0	mA	1-4
E10	SHFT REG CLK	41	0.0	-100.0	-40.0	mA	1-4
E11	MIN INT GAP	49	0.0	-150.0	-60.0	mA	1-4
E12	SER DATA OUT	53	0.0	-225.0	-50.0	mA	1-4
E13	ADAPTER LBK	54	0.0	-150.0	-60.0	mA	1-4
E14	SEL EVEN PAR	55					
E15	INIT DISABLE	56					
E16	B PROTOCOL	57	0.0	-150.0	-60.0	mA	1-4
E17	ENABLE STAT	58	0.0	-225.0	-50.0	mA	1-4
E18	UNLOAD ENBL	66					
E19	LOAD ENABLE	70	0.0	-225.0	-50.0	mA	1-4
E20	LAST WORD	83	0.0	-100.0	-40.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	IOC DATA 15	84	0.0	-150.0	-60.0	mA	1-4
E22	IOC DATA 14	85					
E23	IOC DATA 13	86					
E24	IOC DATA 12	87					
E25	IOC DATA 11	88					
E26	IOC DATA 10	89					
E27	IOC DATA 9	91					
E28	IOC DATA 8	92					
E29	IOC DATA 7	93					
E30	IOC DATA 6	94					
E31	IOC DATA 5	95					
E32	IOC DATA 4	96					
E33	IOC DATA 3	97					
E34	IOC DATA 2	98					
E35	IOC DATA 1	99					
E36	IOC DATA 0	100	0.0	-150.0	-60.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E36
Power supply current limit	2.1 A	NA	E1-E36
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E36
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	E1-E36
low, (VL)	0.0 V	± 100 mV	E1-E36
Output current measurement	NA	± 8.75 mA	E1-E5
	NA	± 0.11 mA	E6
	NA	± 5.75 mA	E7
	NA	± 4.5 mA	E8,E9
	NA	± 3.0 mA	E10
	NA	± 4.5 mA	E11
	NA	± 8.75 mA	E12
	NA	± 4.5 mA	E13-E16
	NA	± 8.75 mA	E17-E19
	NA	± 3.0 mA	E20
	NA	± 4.5 mA	E21-E36

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	EXT CLOCK	2	2.4	-1.55	-0.56	mA	1-3
F2	TERMINATE	4	2.4	-1.75	-0.59	mA	1-3
F3	MASTER CLEAR	7	2.4	-0.6	0.02	mA	1-3
F4	ADAPTER PRES	14	2.4	-4.92	-2.37	mA	1-3
F5	PARAM ENBL	18	2.4	-1.55	-0.59	mA	1-3
F6	CLOCK	24					
F7	SERIAL IN	26					
F8	SHIFT ENABLE	28					
F9	SHIFT CLOCK	30	2.4	-1.55	-0.59	mA	1-3
F10	IOC XMT RCV	34	2.4	-0.6	0.02	mA	1-3
F11	DATA SCAN	35					
F12	CHAIN/INT	36					
F13	IOCA ENABLE	37					
F14	CIII INT EN	38					
F15	T1	43	2.4	-0.6	0.02	mA	1-3
F16	20 MHZ CLK	46	2.4	-2.0	0.05	mA	1-3
F17	19 HZ CLK	47	2.4	-0.4	0.02	mA	1-3
F18	I/O CLK	48	2.4	-4.0	0.1	mA	1-3
F19	SER DATA IN	52	2.4	-2.0	0.05	mA	1-3
F20	SHIFT EN DAT	59	2.4	-3.15	-0.56	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	SER SHFT CLK	61	2.4	-1.55	-0.56	mA	1-3
F22	LAST CLOCK	62	2.4	-3.15	-0.56	mA	1-3
F23	IOC DATA 15	84	2.4	-0.25	0.07	mA	1-3
F24	IOC DATA 14	85					
F25	IOC DATA 13	86					
F26	IOC DATA 12	87					
F27	IOC DATA 11	88					
F28	IOC DATA 10	89					
F29	IOC DATA 9	91					
F30	IOC DATA 8	92					
F31	IOC DATA 7	93					
F32	IOC DATA 6	94					
F33	IOC DATA 5	95					
F34	IOC DATA 4	96					
F35	IOC DATA 3	97					
F36	IOC DATA 2	98					
F37	IOC DATA 1	99					
F38	IOC DATA 0	100	2.4	-0.25	-0.07	mA	1-3

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Check the source to verify that this is the current version before use.

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F38
Power supply current limit	2.1 A	NA	F1-F38
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F38
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	F1-F38
low, (VL)	0.0 V	± 100 mV	F1-F38
Input current measurement	NA	± 49.5 uA	F1
	NA	± 58.0 uA	F2
	NA	± 31.0 uA	F3
	NA	± 127.0 uA	F4
	NA	± 48.0 uA	F5-F9
	NA	± 31.0 uA	F10-F15
	NA	± 102.0 uA	F16
	NA	± 21.0 uA	F17
	NA	± 205.0 uA	F18
	NA	± 102.0 uA	F19
	NA	± 129.0 uA	F20
	NA	± 49.5 uA	F21
	NA	± 129.0 uA	F22
	NA	± 16.0 uA	F23-F38

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	EXT CLOCK	2	0.4	-1.55	-0.56	mA	1-3
G2	TERMINATE	4	0.4	-1.75	-0.59	mA	1-3
G3	MASTER CLEAR	7	0.4	-0.6	0.02	mA	1-3
G4	ADAPTER PRES	14	0.4	-4.92	-2.37	mA	1-3
G5	PARAM ENBL	18	0.4	-1.55	-0.59	mA	1-3
G6	CLOCK	24					
G7	SERIAL IN	26					
G8	SHIFT ENABLE	28					
G9	SHIFT CLOCK	30	0.4	-1.55	-0.59	mA	1-3
G10	IOC XMT RCV	34	0.4	-0.6	0.02	mA	1-3
G11	DATA SCAN	35					
G12	CHAIN/INT	36					
G13	IOCA ENABLE	37					
G14	CIII INT EN	38					
G15	T1	43	0.4	-0.6	0.02	mA	1-3
G16	20 MHZ CLK	46	0.4	-2.0	0.05	mA	1-3
G17	19 HZ CLK	47	0.4	-0.4	0.02	mA	1-3
G18	I/O CLK	48	0.4	-4.0	0.1	mA	1-3
G19	SER DATA IN	52	0.4	-2.0	0.05	mA	1-3
G20	SHIFT EN DAT	59	0.4	-3.15	-0.56	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	SER SHFT CLK	61	0.4	-1.55	-0.56	mA	1-3
G22	LAST CLOCK	62	0.4	-3.15	-0.56	mA	1-3
G23	IOC DATA 15	84	0.4	-0.25	0.07	mA	1-3
G24	IOC DATA 14	85					
G25	IOC DATA 13	86					
G26	IOC DATA 12	87					
G27	IOC DATA 11	88					
G28	IOC DATA 10	89					
G29	IOC DATA 9	91					
G30	IOC DATA 8	92					
G31	IOC DATA 7	93					
G32	IOC DATA 6	94					
G33	IOC DATA 5	95					
G34	IOC DATA 4	96					
G35	IOC DATA 3	97					
G36	IOC DATA 2	98					
G37	IOC DATA 1	99					
G38	IOC DATA 0	100	0.4	-0.25	0.07	mA	1-3

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NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G38
Power supply current limit	2.1 A	NA	G1-G38
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G38
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	G1-G38
low, (VL)	0.0 V	± 100 mV	G1-G38
Input current measurement	NA	± 49.5 uA	G1
	NA	± 58.0 uA	G2
	NA	± 31.0 uA	G3
	NA	± 127.0 uA	G4
	NA	± 48.0 uA	G5-G9
	NA	± 31.0 uA	G10-G15
	NA	± 102.0 uA	G16
	NA	± 21.0 uA	G17
	NA	± 205.0 uA	G18
	NA	± 102.0 uA	G19
	NA	± 129.0 uA	G20
	NA	± 49.5 uA	G21
	NA	± 129.0 uA	G22
	NA	± 16.0 uA	G23-G38

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	SER CLKOUT B	3	-0.4	2.4	4.5	V	1-4
H2	IOCA CLR	5					
H3	SER CLKOUT A	6					
H4	10 MHZ A	13					
H5	10 MHZ B	15	-0.4	2.4	4.5	V	1-4
H6	SERIAL OUT	19	-0.16	2.4	4.5	V	1-4
H7	CHAN REQ	23	-0.4	2.4	4.5	V	1-4
H8	PARAM OUT	27					
H9	SHFT REG CLK	41					
H10	MIN INT GAP	49					
H11	SER DATA OUT	53					
H12	ADAPTER LBK	54					
H13	SEL EVEN PAR	55					
H14	INIT DISABLE	56					
H15	B PROTOCOL	57					
H16	ENABLE STAT	58					
H17	UNLOAD ENBL	66					
H18	LOAD ENABLE	70					
H19	LAST WORD	83					
H20	IOC DATA 15	84	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H21	IOC DATA 14	85	-0.4	2.4	4.5	V	1-4
H22	IOC DATA 13	86					
H23	IOC DATA 12	87					
H24	IOC DATA 11	88					
H25	IOC DATA 10	89					
H26	IOC DATA 9	91					
H27	IOC DATA 8	92					
H28	IOC DATA 7	93					
H29	IOC DATA 6	94					
H30	IOC DATA 5	95					
H31	IOC DATA 4	96					
H32	IOC DATA 3	97					
H33	IOC DATA 2	98					
H34	IOC DATA 1	99					
H35	IOC DATA 0	100	-0.4	2.4	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H35
Power supply current limit	2.1 A	NA	H1-H35
Forcing current, (Ioh)	-0.4 mA -0.16 mA -0.4 mA	± 40 uA ± 16 uA ± 40 uA	H1-H5 H6 H7-H35
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H35 H1-H35
Output voltage measurement	NA	± 105 mV	H1-H35

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

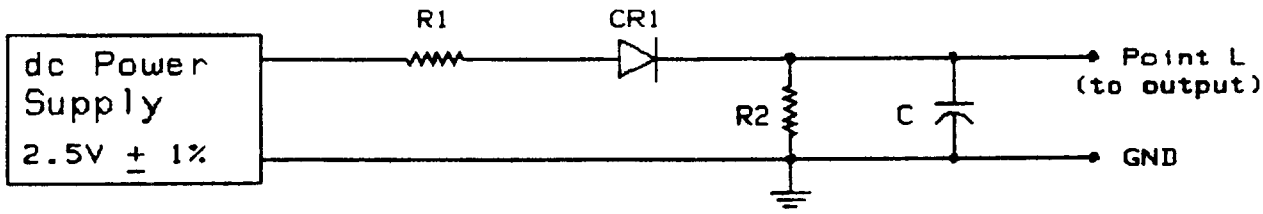
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
3,5,6,13,15,23, 27,41,49,53,54, 55,56,57,58,66, 70,83,84,85,86, 87,88,89,91,92, 93,94,95,96,97, 98,99,100	6200	6.0	0.5
16,32	NA	1.6	0.5
19	16000	1.6	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J1	SER CLKOUT B	3	6.0	0.0	0.5	V	1-4
J2	IOCA CLR	5					
J3	SER CLKOUT A	6					
J4	10 MHZ A	13					
J5	10 MHZ B	15	6.0	0.0	0.5	V	1-4
J6	TERM MODE	16	1.6	0.0	0.5	V	1-4
J7	SERIAL OUT	19	1.6	0.0	0.5	V	1-4
J8	CHAN REQ	23	6.0	0.0	0.5	V	1-4
J9	PARAM OUT	27	6.0	0.0	0.5	V	1-4
J10	CHAN PRES	32	1.6	0.0	0.5	V	1-4
J11	SHFT REG CLK	41	6.0	0.0	0.5	V	1-4
J12	MIN INT GAP	49					
J13	SER DATA OUT	53					
J14	ADAPTER LBK	54					
J15	SEL EVEN PAR	55					
J16	INIT DISABLE	56					
J17	B PROTOCOL	57					
J18	ENABLE STAT	58					
J19	UNLOAD ENBL	66					
J20	LOAD ENABLE	70	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J21	LAST WORD	83	6.0	0.0	0.5	V	1-4
J22	IOC DATA 15	84					
J23	IOC DATA 14	85					
J24	IOC DATA 13	86					
J25	IOC DATA 12	87					
J26	IOC DATA 11	88					
J27	IOC DATA 10	89					
J28	IOC DATA 9	91					
J29	IOC DATA 8	92					
J30	IOC DATA 7	93					
J31	IOC DATA 6	94					
J32	IOC DATA 5	95					
J33	IOC DATA 4	96					
J34	IOC DATA 3	97					
J35	IOC DATA 2	98					
J36	IOC DATA 1	99					
J37	IOC DATA 0	100	6.0	0.0	0.5	V	1-4

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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J37
Power supply current limit	2.1 A	NA	J1-J37
Forcing current, (Iol)	6.0 mA 1.6 mA 6.0 mA 1.6 mA 6.0 mA	± 600 uA ± 160 uA ± 600 uA ± 160 uA ± 600 uA	J1-J5 J6,J7 J8,J9 J10 J11-J37
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	J1-J37 J1-J37
Output voltage measurement	NA	± 25 mV	J1-J37

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high and low level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
High level							
K1	CHAN PRES	32	2.4	0.0	0.25	mA	1-3
K2	ENABLE STAT	58	2.4	-0.05	0.05	mA	1-3
K3	UNLOAD ENBL	66	〃	〃	〃	〃	〃
K4	LOAD ENABLE	70	2.4	-0.05	0.05	mA	1-3
Low level							
K5	ENABLE STAT	58	0.4	-0.05	0.05	mA	1-3
K6	UNLOAD ENBL	66	〃	〃	〃	〃	〃
K7	LOAD ENABLE	70	0.4	-0.05	0.05	mA	1-3

NOTES:

1. Output leakage current, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K1-K7
Power supply current limit	2.1 A	NA	K1-K7
Forcing voltage, (Voh)	2.4 V	± 50 mV	K1-K4
(Vol)	0.4 V	± 50 mV	K5-K7
Pattern voltage high, (VH)	3.0 V	± 100 mV	K1-K7
low, (VL)	0.0 V	± 100 mV	K1-K7

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output current measurement	NA NA	± 12.5 uA ± 5.0 uA	K1 K2-K7

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	MASTER CLEAR	7	-18.0	-1.2	0.0	V	1-3
L2	IOC XMT RCV	34					
L3	DATA SCAN	35					
L4	CHAIN/INT	36					
L5	IOCA ENABLE	37					
L6	CIII INT EN	38					
L7	T1	43					
L8	20 MHZ CLK	46	-18.0	-1.2	0.0	V	1-3
L9	19 HZ CLK	47	-18.0	-1.5	0.0	V	1-3
L10	I/O CLK	48	-18.0	-1.2	0.0	V	1-3
L11	SER DATA IN	52	-18.0	-1.2	0.0	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L11
Power supply current limit	NA	NA	L1-L11
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L11
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L11 L1-L11
Input voltage measurement	 NA NA NA	 ± 60 mV ± 75 mV ± 60 mV	 L1-L8 L9 L10,L11

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M	PSFC	1	8.0	15.0	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M
Capacitance measurement	NA	± 0.35 uF	M

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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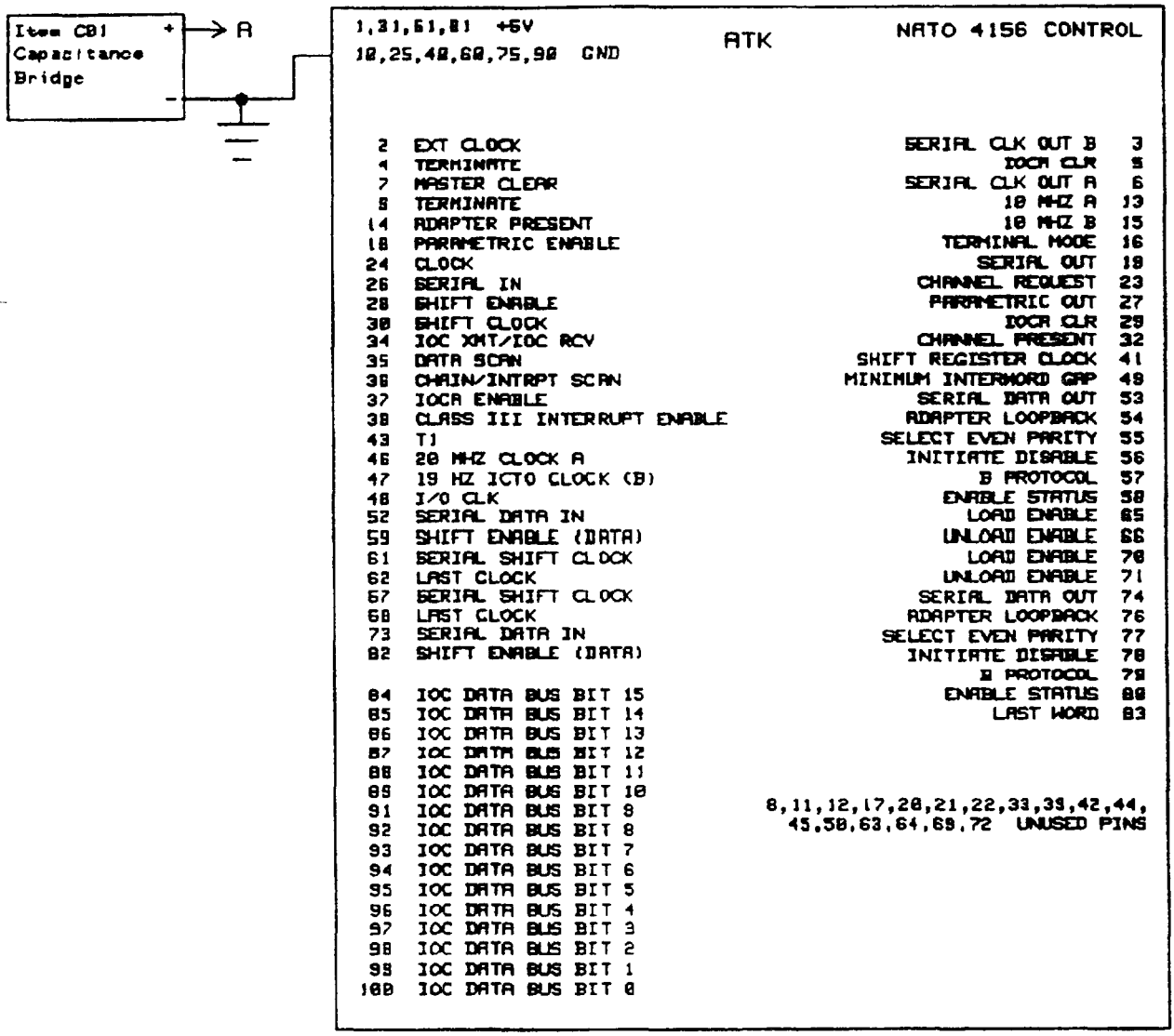


FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31 51,81	582	1398	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	2.1 A	NA	N
Pattern voltage high, (VH)	3.0 V	± 100 mV	N
low, (VL)	0.0 V	± 100 mV	N
Power supply current measurement	NA	± 40.8 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (300 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (300 nS)	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	2.1 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 20 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

3. Abbreviations used are defined in 6.1.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (300 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (300 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (300 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (300 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	2.1 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

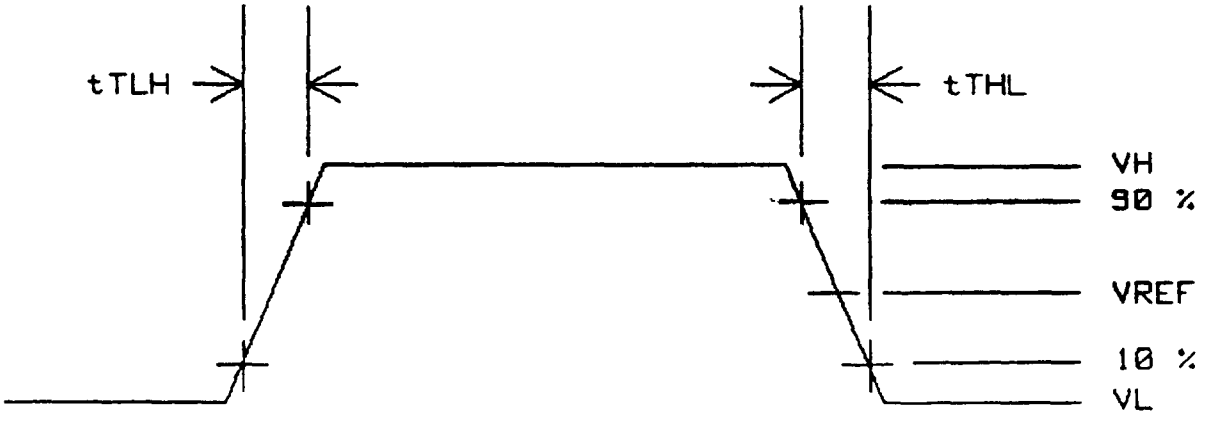
(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 11. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	MASTER CLEAR to IOCA CLR	-	-	-	55	55	55	nS	11
2	tpd	I/O CLK to TERM MODE	-	-	-	109	109	109	nS	11
3	tpd	DATA SCAN to CHAN REQ	-	-	-	48	48	48	nS	11
4	tpd	20 MHZ CLK to LOAD ENABLE	-	-	-	104	104	104	nS	11
5	tpd	I/O CLK to INIT DISABLE	-	-	-	57	57	57	nS	11
6	tpd	I/O CLK to IOC DATA 0	-	-	-	47	47	47	nS	11
7	tpd	SER SHFT CLK to SER DATA OUT	-	-	-	58	58	58	nS	11

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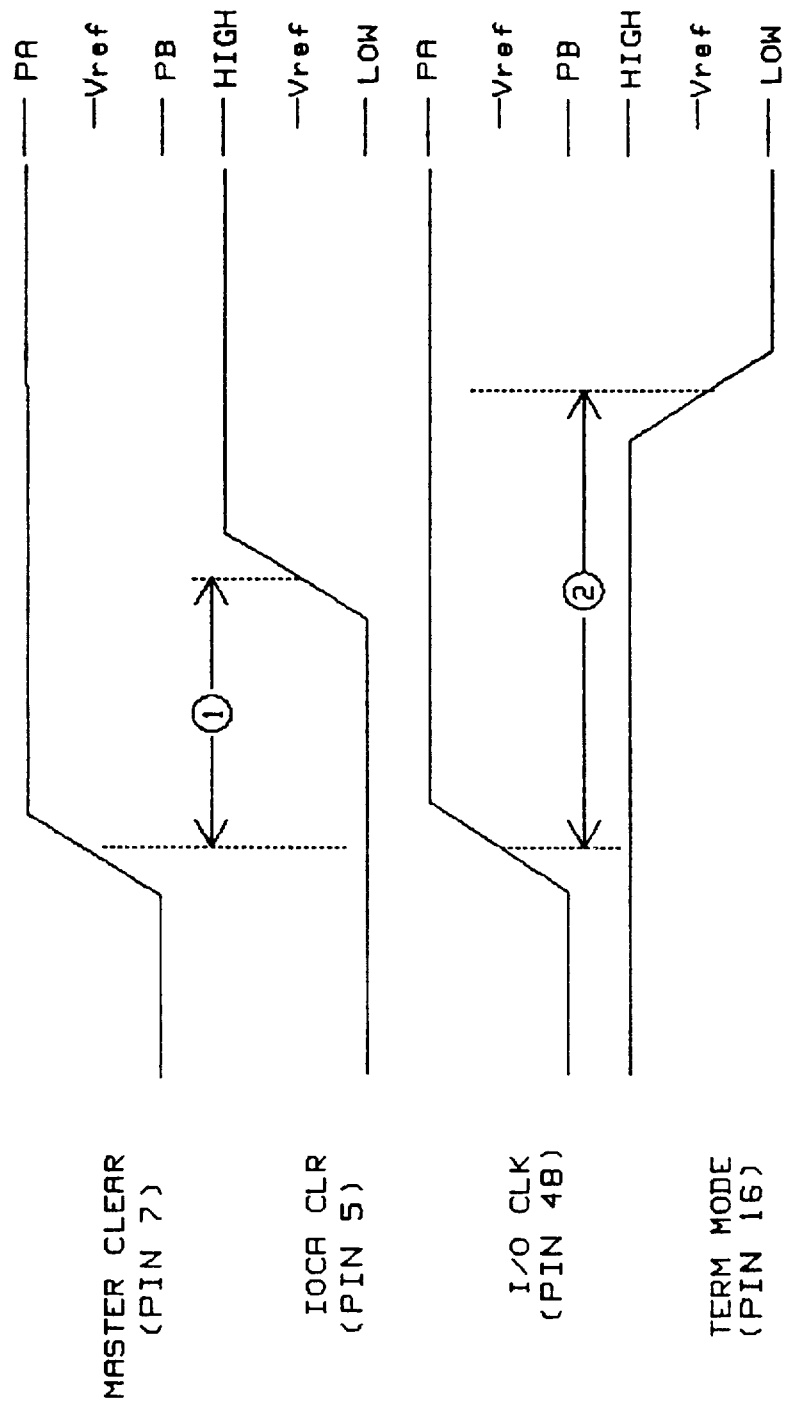


FIGURE 11a. Dynamic waveforms for table XVII.

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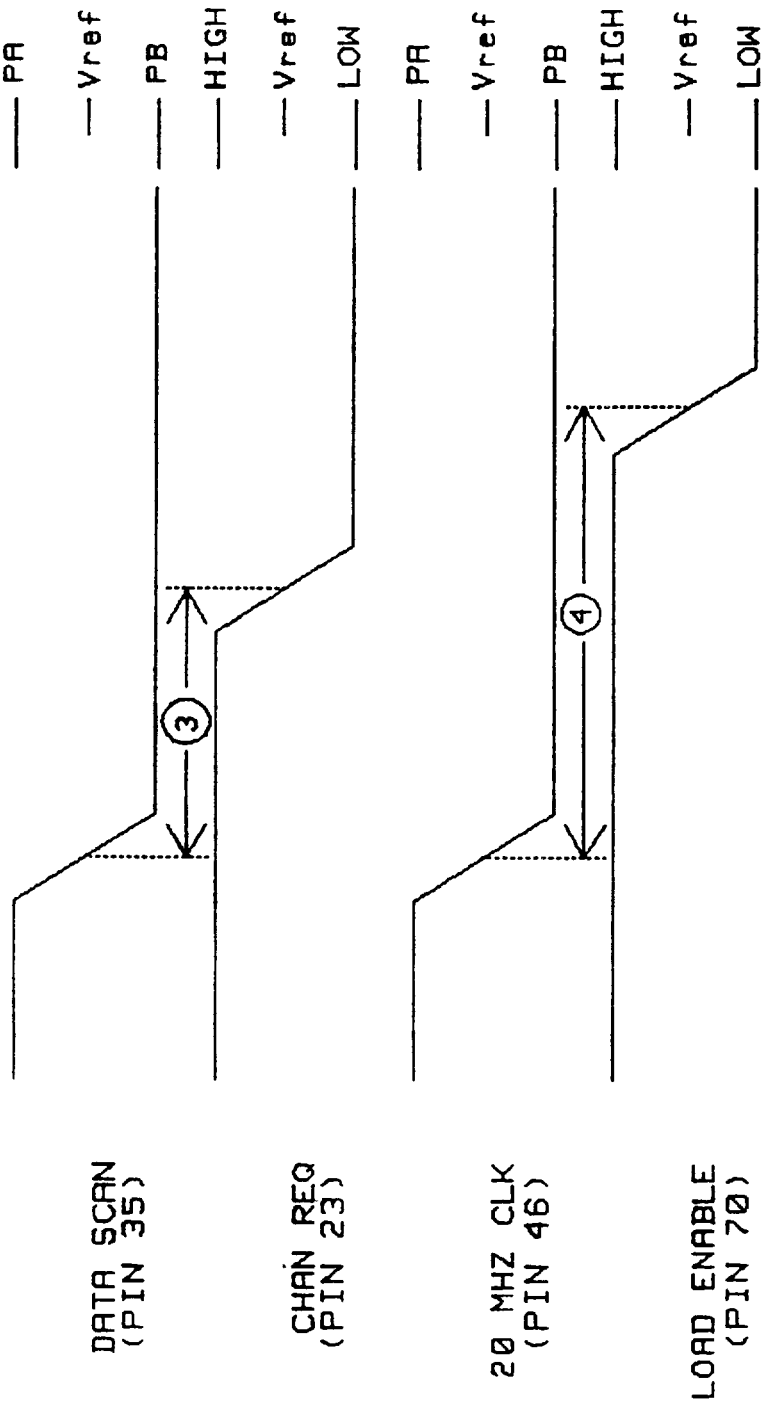


FIGURE 11b. Dynamic waveforms for table XVII.

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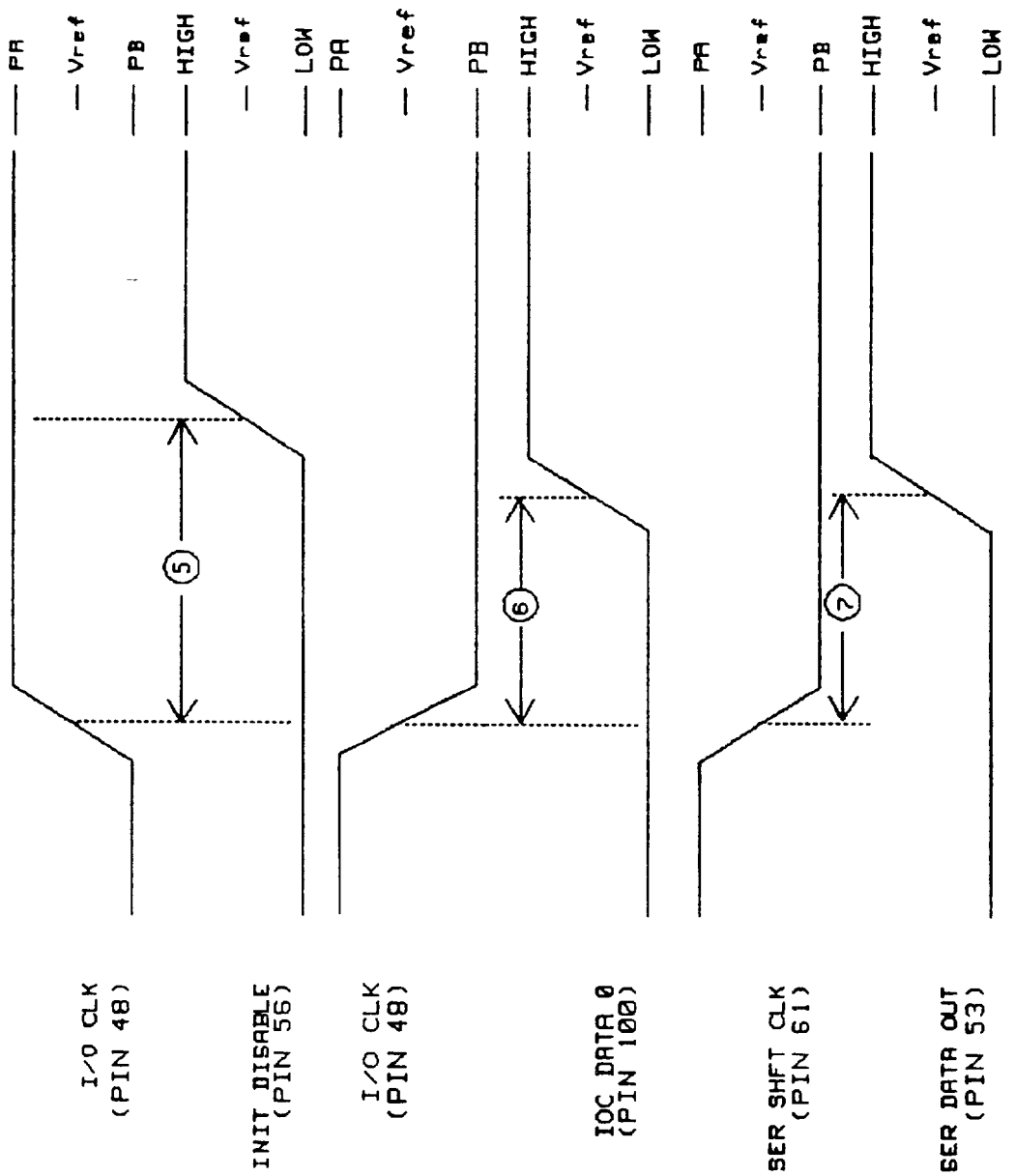


FIGURE 11c. Dynamic waveforms for table XVII.

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TABLE XVIII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Inputs						
R1	EXT CLOCK	2	2	100	pF	1-4
R2	TERMINATE	4	4	100	pF	1-4
R3	MASTER CLEAR	7	2	200	pF	1-4
R4	TERMINATE	9	4	100	pF	1-4
R5	ADAPTER PRES	14	14	100	pF	1-4
R6	PARAM ENBL	18	4	75	pF	1-4
R7	CLOCK	24	4	50	pF	1-4
R8	SERIAL IN	26	4	75	pF	1-4
R9	SHIFT ENABLE	28				
R10	SHIFT CLOCK	30	4	75	pF	1-4
R11	IOC XMT RCV	34	2	100	pF	1-4
R12	DATA SCAN	35				
R13	CHAIN/INT	36				
R14	IOCA ENABLE	37				
R15	CIII INT EN	38				
R16	T1	43				
R17	20 MHZ CLK	46	2	100	pF	1-4
R18	19 HZ CLK	47	2	75	pF	1-4
R19	I/O CLK	48	4	100	pF	1-4
R20	SER DATA IN	52	2	125	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
	Output					
R21	SER DATA OUT	53	2	100	pF	1-4
	Inputs					
R22	SHIFT EN DAT	59	4	125	pF	1-4
R23	SER SHFT CLK	61	4	100	pF	1-4
R24	LAST CLOCK	62				
R25	SER SHFT CLK	67				
R26	LAST CLOCK	68	4	100	pF	1-4
R27	SER DATA IN	73	2	125	pF	1-4
R28	SHIFT EN DAT	82	4	125	pF	1-4
	Bidirectionals					
R29	IOC DATA 15	84	4	75	pF	1-4
R30	IOC DATA 14	85				
R31	IOC DATA 13	86				
R32	IOC DATA 12	87				
R33	IOC DATA 11	88				
R34	IOC DATA 10	89				
R35	IOC DATA 9	91				
R36	IOC DATA 8	92				
R37	IOC DATA 7	93				
R38	IOC DATA 6	94				
R39	IOC DATA 5	95	4	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R40	IOC DATA 4	96	4	75	pF	1-4
R41	IOC DATA 3	97				
R42	IOC DATA 2	98				
R43	IOC DATA 1	99				
R44	IOC DATA 0	100	4	75	pF	1-4

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NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R44
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R44
Capacitance measurement	NA	± 4.9 pF	R1
	NA	± 4.8 pF	R2
	NA	± 9.9 pF	R3
	NA	± 4.8 pF	R4
	NA	± 4.3 pF	R5
	NA	± 3.55 pF	R6
	NA	± 2.3 pF	R7
	NA	± 3.55 pF	R8-R10
	NA	± 4.9 pF	R11-R17
	NA	± 3.65 pF	R18
	NA	± 4.8 pF	R19
	NA	± 6.15 pF	R20
	NA	± 4.9 pF	R21
	NA	± 6.05 pF	R22
	NA	± 4.8 pF	R23-R26
	NA	± 6.15 pF	R27
	NA	± 6.05 pF	R28
	NA	± 3.55 pF	R29-R44

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 12.

4. Abbreviations used are defined in 6.1.

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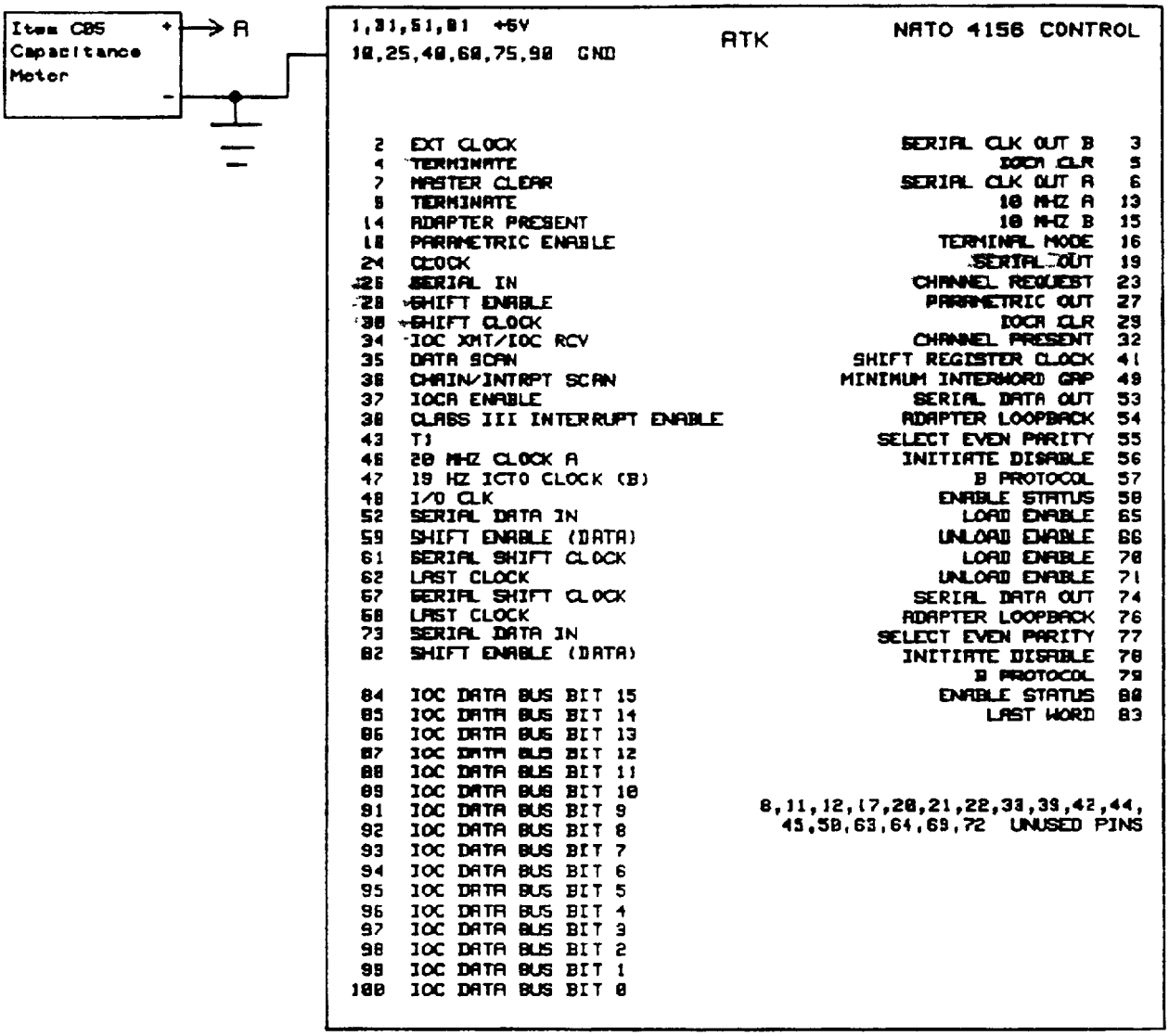


FIGURE 12. Input and output capacitance test circuit for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T	PSTA	1	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T
Power supply current limit	2.1 A	NA	T
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T T
Power supply transient amplitude measurement	NA	± 20 mV	T

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 13 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 14.

4. Abbreviations used are defined in 6.1.

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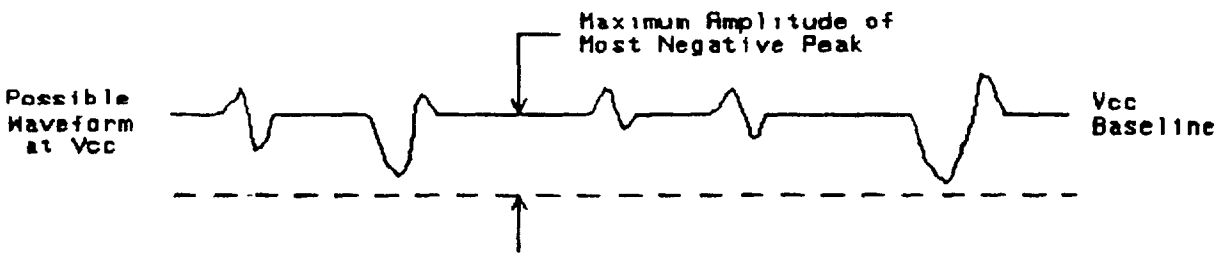
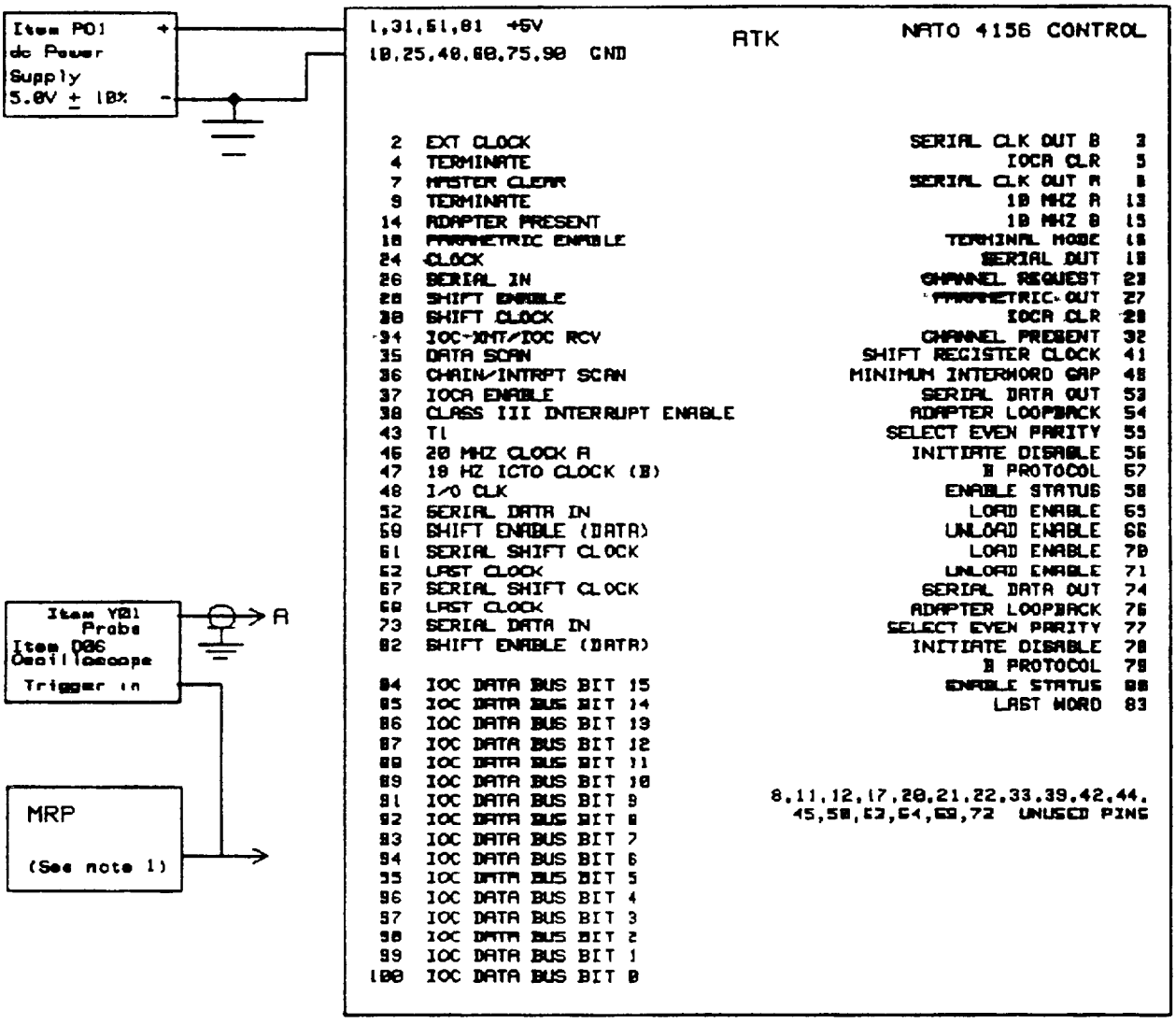


FIGURE 13. Power supply transient amplitude waveform for table XIX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 14. Power supply transient amplitude test circuit for table XIX.

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER	2 DOCUMENT DATE (YYMMDD)
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3 DOCUMENT TITLE: Modules, Standard Electronic, NATO-STD-4156 (Modified) Channel Contro' Module, Key Code ATK

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a NAME Technical Point of Contact (TPOC): Mr. Robert F. Dilger (Code 602A) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON TPOC: 812-854-1528 88-482-1528
c ADDRESS (Include Zip Code) Commanding Officer Naval Weapons Support Center ATTN: Robert F. Dilger, Bldg. 2940W, Code 602A Crane, Indiana 47522-5060	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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